**SOC Lab INT-001: Build the Virtual Lab Foundations**

## Lab Environment & Tools

- **Hypervisor:** VMware Workstation Pro
- **Host System:** Physical machine with virtualization enabled (VT-x / AMD-V)
- **Operating Systems:**
  - Windows 11 (Client VM)
  - Linux (Server VM)

## Lab Setup Overview

The lab environment consists of two virtual machines connected via:

- **NAT Network Adapter** → Internet access
- **Host-only/Internal Adapter** → VM-to-VM communication

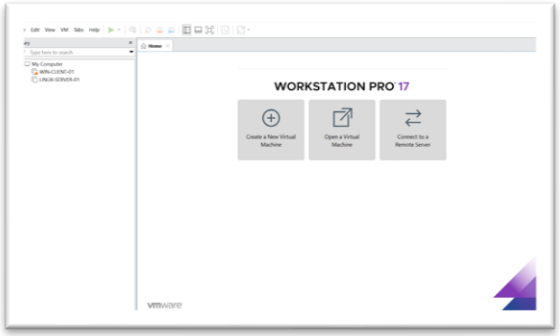
This dual-network setup allows:

- Controlled internal communication
- Safe internet access for updates and tools
- Isolation from the host network

## Implementation Steps

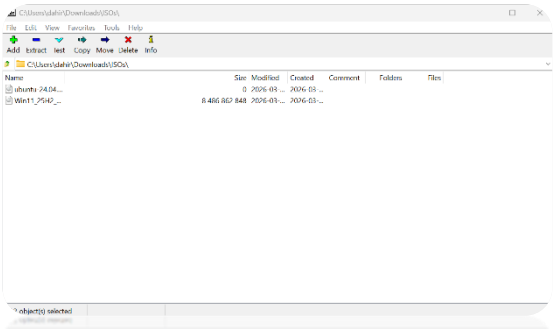
### 1. Hypervisor Installation & Preparation

- Installed **VMware Workstation Pro**
- Verified CPU virtualization support in BIOS/UEFI
- Created an organized **folder structure** for lab resources



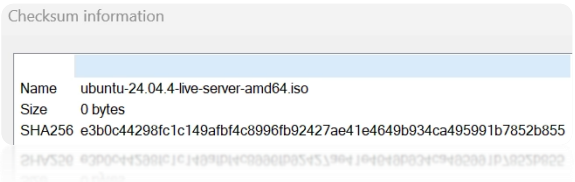
## 2. ISO Acquisition & Verification

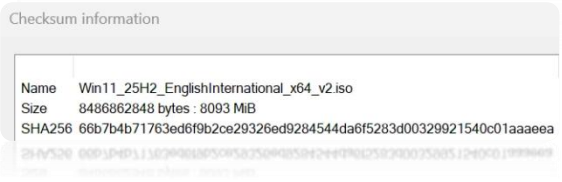
- Downloaded Windows 11 ISO securely (incognito mode)
- Verified integrity using:
    - **SHA-256 checksum via 7-Zip**



This ensures:

- No tampering
- Safe installation media



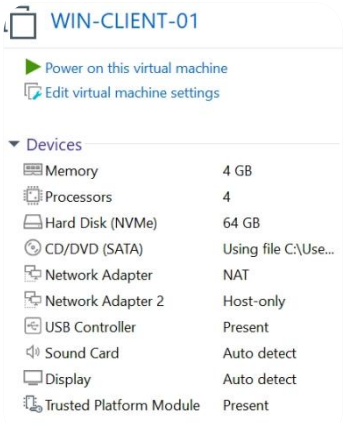

## 3. Virtual Machine Creation

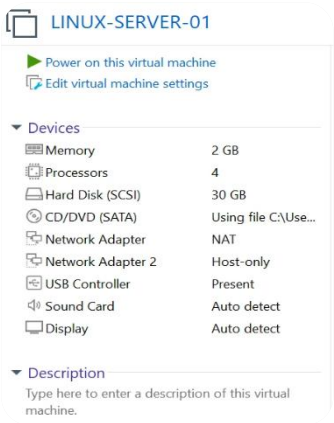
### Windows VM (WIN-CLIENT-01)

- CPU: 2 cores
- RAM: 4 GB
- Disk: 60 GB (dynamic allocation)

### Linux VM (LINUX-SERVER-01)

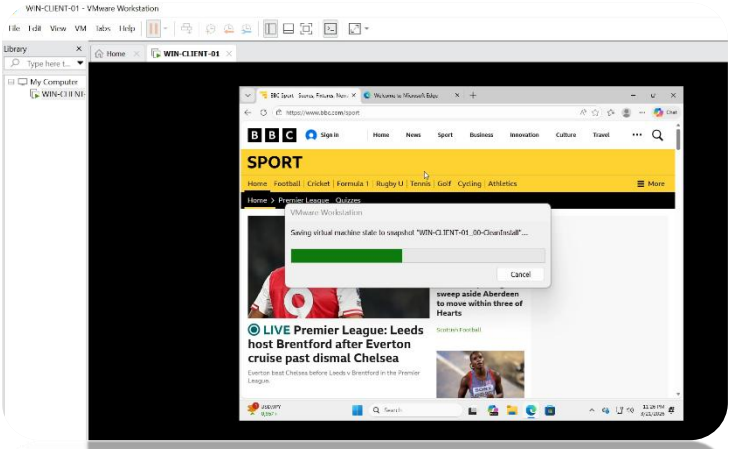
- Installed and configured similarly

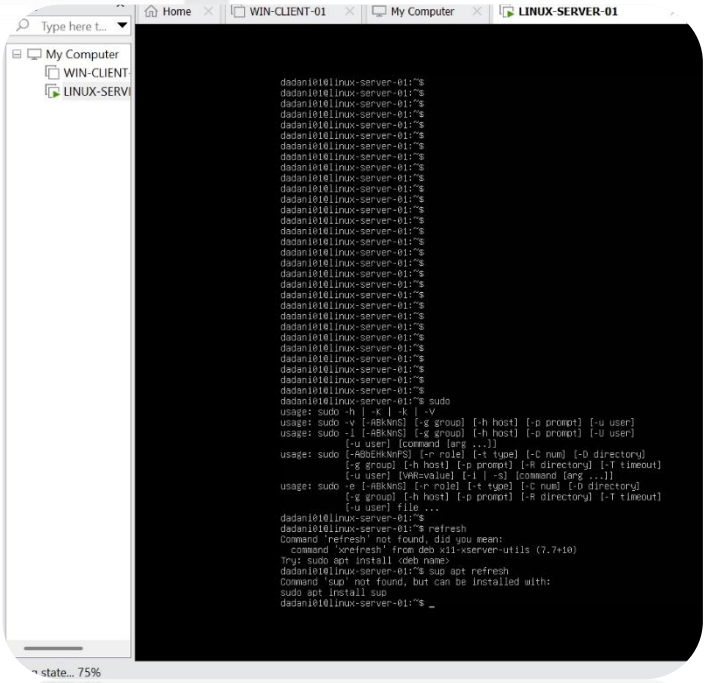


## 4. Snapshot Management

- Created **initial snapshots** for both VMs
- Used snapshots to:
  - Save clean states
  - Enable rollback after testing or misconfiguration



## 5. Network Configuration

Each VM was configured with **two adapters**:

| Adapter | Purpose |
|---|---|
| NAT | Internet access |
| Internal / Host-only | Communication between VMs |

```
:\Users\gulf->ipconfig

indows IP Configuration


thernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : localdomain
   Link-local IPv6 Address . . . . . : fe80::99ac:a4d7:b9c5:c89f%3
   IPv4 Address. . . . . . . . . . . : 192.168.181.128
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.181.2

thernet adapter Ethernet1:

   Connection-specific DNS Suffix  . : localdomain
   Link-local IPv6 Address . . . . . : fe80::2f1:dc9a:96f3:f8f4%14
   IPv4 Address. . . . . . . . . . . : 192.168.120.129
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

:\Users\gulf->
```

```
dadani01@linux-server-01:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:b3:81:93 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.181.129/24 metric 100 brd 192.168.181.255 scope global dynamic ens33
       valid_lft 1295sec preferred_lft 1295sec
    inet6 fe80::20c:29ff:feb3:8193/64 scope link
       valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:b3:81:9d brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 192.168.120.130/24 brd 192.168.120.255 scope global ens37
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:feb3:819d/64 scope link
       valid_lft forever preferred_lft forever
dadani01@linux-server-01:~$
```

## 6. Connectivity Testing

- Configured static IP on Linux internal interface:

sudo nano /etc/netplan/50-cloud-init.yaml

Example:

```
ens37:
  dhcp4: false
  addresses:
  - 192.168.120.130/24
```

- Applied configuration:

sudo netplan apply

- Verified connectivity using:

ip a
ping <target>

```
dadani01@linux-server-01:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
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    altname enp2s1
    inet 192.168.181.129/24 metric 100 brd 192.168.181.255 scope global dynamic ens33
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       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:feb3:819d/64 scope link
       valid_lft forever preferred_lft forever
dadani01@linux-server-01:~$
```

## 7. Firewall Configuration (Windows)

Initially:

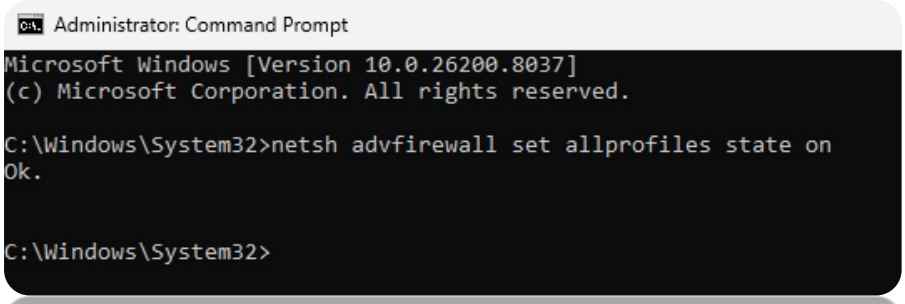
netsh advfirewall set allprofiles state off

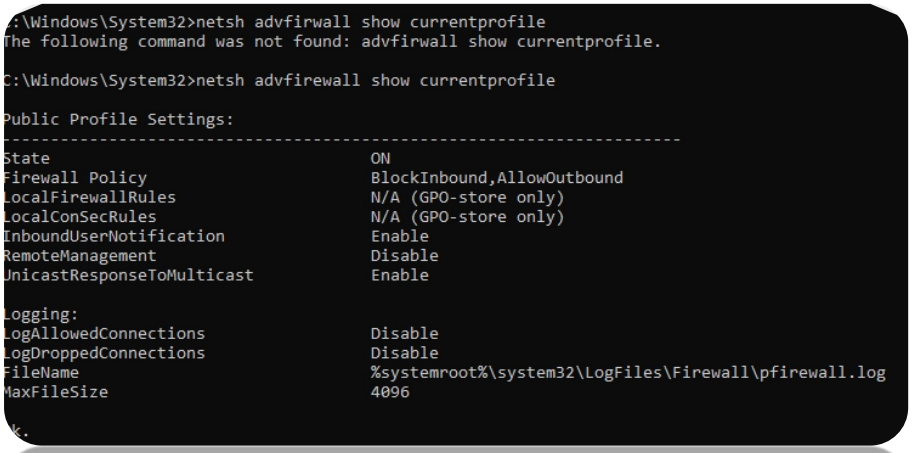
Then re-enabled securely:

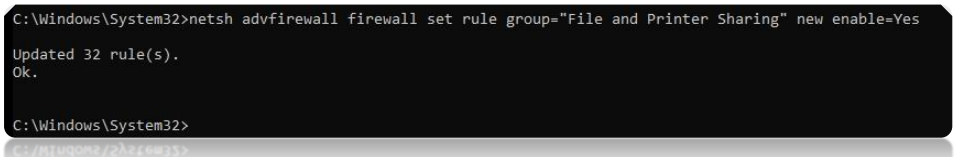
netsh advfirewall set allprofiles state on

Allowed ICMP (ping):

netsh advfirewall firewall add rule name="Allow ICMPv4-In Public" dir=in action=allow protocol=icmpv4:8,any profile=public



# SOC Lab INT-001: Build the Virtual Lab Foundations

```
Windows IP Configuration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Ethernet adapter Ethernet1:

   Connection-specific DNS Suffix  . : localdomain
   Link-local IPv6 Address . . . . . : fe80::2f1:dc9a:96f3:f8f4%14
   IPv4 Address. . . . . . . . . . . : 192.168.120.129
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

```
      valid_lft forever preferred_lft forever
dadani01@linux-server-01:~$ ping 192.168.181.128
PING 192.168.181.128 (192.168.181.128) 56(84) bytes of data.
64 bytes from 192.168.181.128: icmp_seq=1 ttl=128 time=1.26 ms
64 bytes from 192.168.181.128: icmp_seq=2 ttl=128 time=0.995 ms
64 bytes from 192.168.181.128: icmp_seq=3 ttl=128 time=0.884 ms
64 bytes from 192.168.181.128: icmp_seq=4 ttl=128 time=0.813 ms
64 bytes from 192.168.181.128: icmp_seq=5 ttl=128 time=0.596 ms
64 bytes from 192.168.181.128: icmp_seq=6 ttl=128 time=0.695 ms
64 bytes from 192.168.181.128: icmp_seq=7 ttl=128 time=0.656 ms
64 bytes from 192.168.181.128: icmp_seq=8 ttl=128 time=1.15 ms
64 bytes from 192.168.181.128: icmp_seq=9 ttl=128 time=1.02 ms
64 bytes from 192.168.181.128: icmp_seq=10 ttl=128 time=0.637 ms
64 bytes from 192.168.181.128: icmp_seq=11 ttl=128 time=0.399 ms
64 bytes from 192.168.181.128: icmp_seq=12 ttl=128 time=0.802 ms

64 bytes from 192.168.181.128: icmp_seq=13 ttl=128 time=0.664 ms
64 bytes from 192.168.181.128: icmp_seq=14 ttl=128 time=0.526 ms
64 bytes from 192.168.181.128: icmp_seq=15 ttl=128 time=0.567 ms
^C
--- 192.168.181.128 ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 14066ms
rtt min/avg/max/mdev = 0.399/0.778/1.260/0.235 ms
dadani01@linux-server-01:~$ ping 192.168.120.129
PING 192.168.120.129 (192.168.120.129) 56(84) bytes of data.
64 bytes from 192.168.120.129: icmp_seq=1 ttl=128 time=1.05 ms
64 bytes from 192.168.120.129: icmp_seq=2 ttl=128 time=0.949 ms
64 bytes from 192.168.120.129: icmp_seq=3 ttl=128 time=0.824 ms
64 bytes from 192.168.120.129: icmp_seq=4 ttl=128 time=0.659 ms
64 bytes from 192.168.120.129: icmp_seq=5 ttl=128 time=0.750 ms
64 bytes from 192.168.120.129: icmp_seq=6 ttl=128 time=0.932 ms
64 bytes from 192.168.120.129: icmp_seq=7 ttl=128 time=0.614 ms
64 bytes from 192.168.120.129: icmp_seq=8 ttl=128 time=0.604 ms
64 bytes from 192.168.120.129: icmp_seq=9 ttl=128 time=0.546 ms
^C
--- 192.168.120.129 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8040ms
rtt min/avg/max/mdev = 0.546/0.770/1.053/0.168 ms
dadani01@linux-server-01:~$ ping 192.168.120.130
PING 192.168.120.130 (192.168.120.130) 56(84) bytes of data.
64 bytes from 192.168.120.130: icmp_seq=1 ttl=64 time=0.051 ms
64 bytes from 192.168.120.130: icmp_seq=2 ttl=64 time=0.055 ms
64 bytes from 192.168.120.130: icmp_seq=3 ttl=64 time=0.052 ms
64 bytes from 192.168.120.130: icmp_seq=4 ttl=64 time=0.056 ms
^C
--- 192.168.120.130 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3052ms
rtt min/avg/max/mdev = 0.051/0.053/0.056/0.002 ms
```

## Security Hardening

### Windows Hardening

- Disabled Guest account:

net user guest /active:no

- Enabled Microsoft Defender:
    - Real-time protection
    - Cloud-delivered protection
    - Tamper protection
- Disabled insecure services:

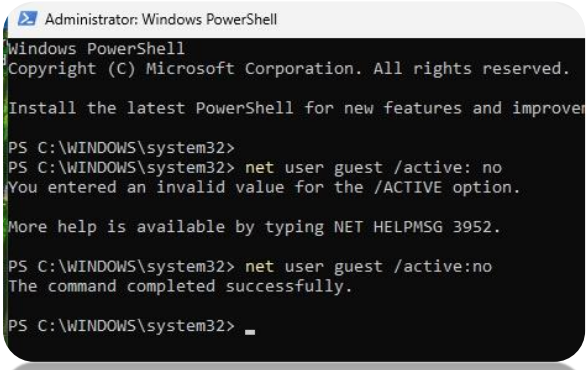
Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol

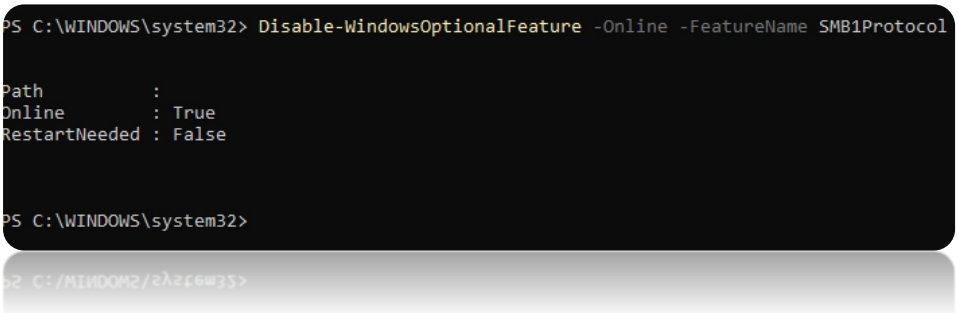
- Enabled auditing:

auditpol /set /category:* /success:enable /failure:enable

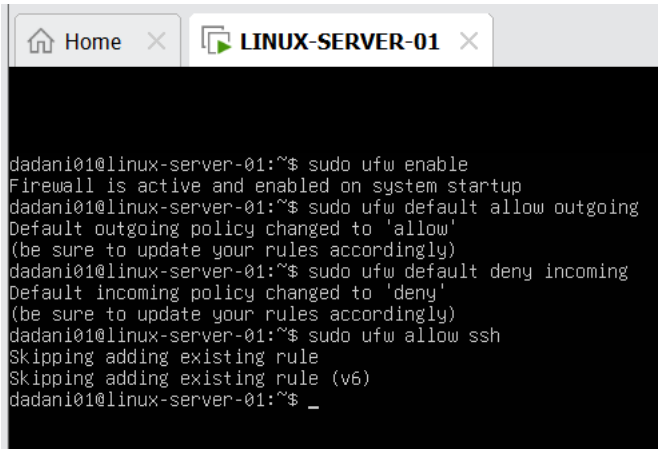
- Verified firewall rules:

netsh advfirewall firewall show rule name=all



## Linux Hardening




- Enabled firewall:

sudo ufw enable
sudo ufw default deny incoming
sudo ufw default allow outgoing

- Allowed SSH:

sudo ufw allow ssh

- Disabled root login:

sudo nano /etc/ssh/sshd_config
# Set:
PermitRootLogin no

- Restarted SSH:

sudo systemctl restart ssh

- Installed security tools:

sudo apt install fail2ban auditd

- Updated system:

sudo apt update && sudo apt upgrade -y

## Monitoring & Logging

- Linux logs:
    - /var/log/auth.log
- File permissions:

ls -l

- Service management:

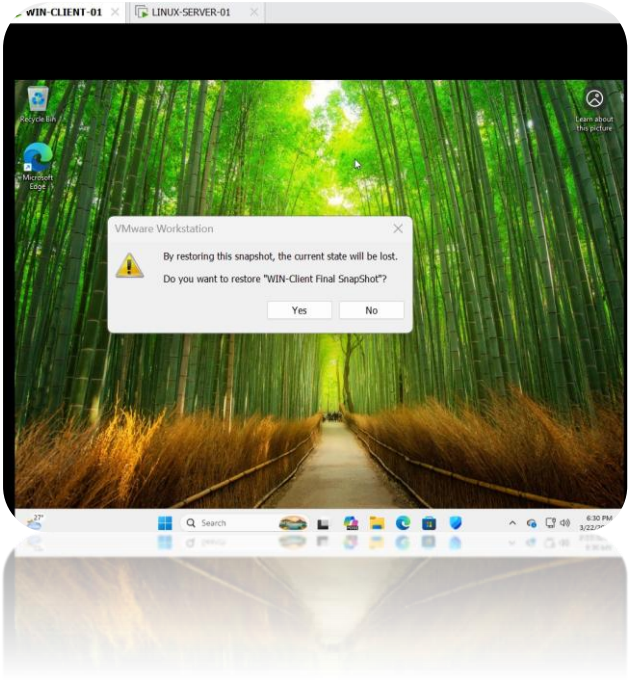
systemctl list-unit-files --type=service

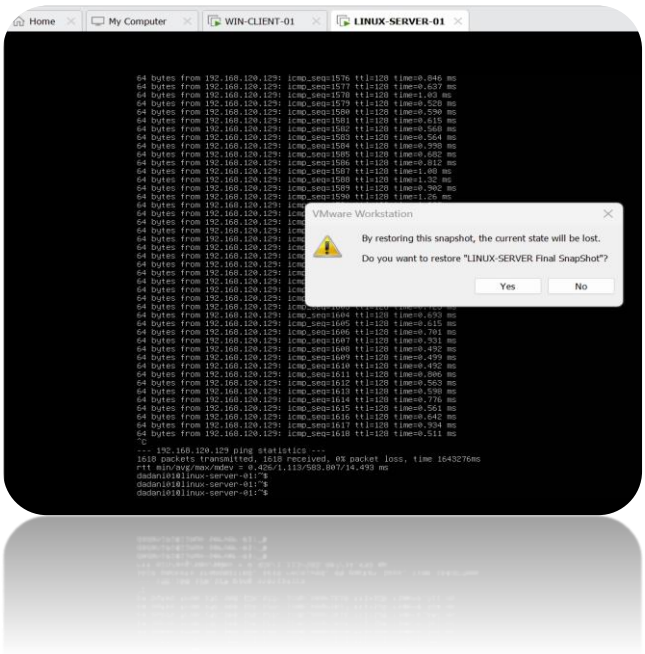
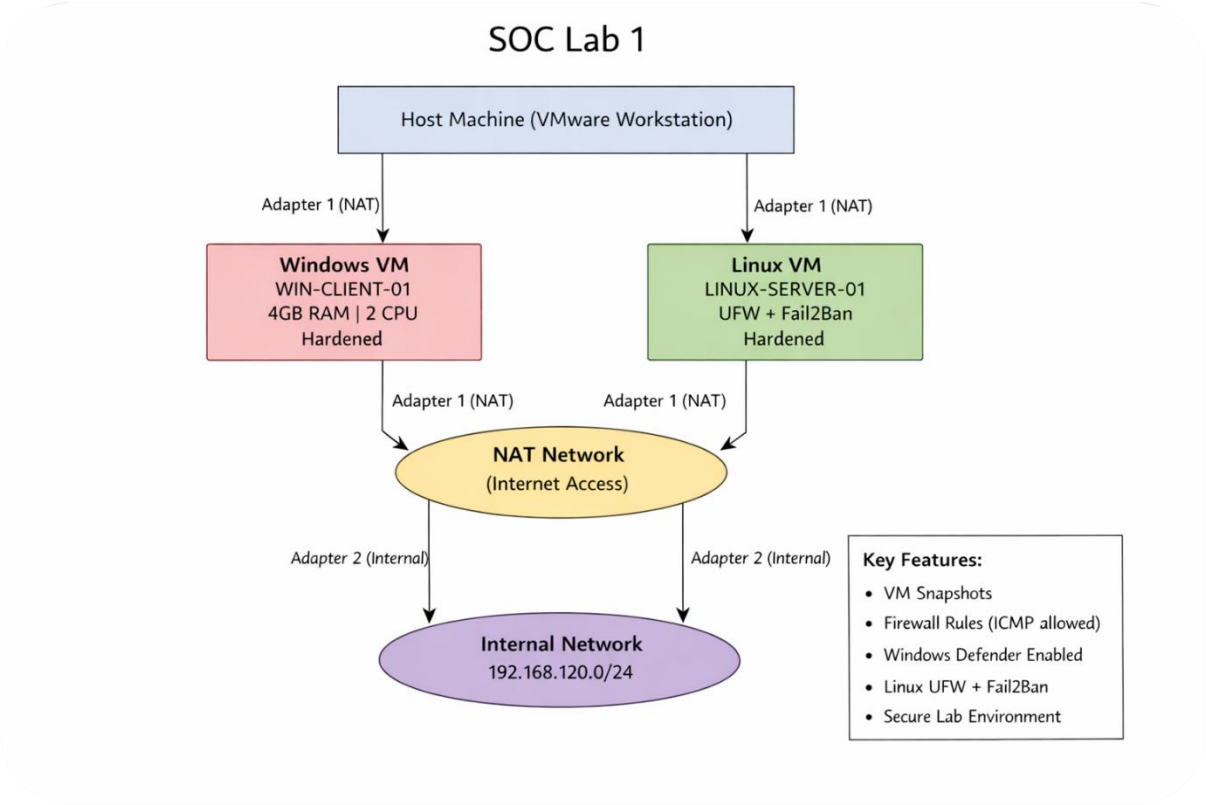
## Snapshot & Recovery

- Reverted to previous snapshots to test recovery
- Created **final baseline snapshots** after hardening

This provides:

- A stable starting point for future labs
- Easy rollback for experiments

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